

FORM TP 2016273



TEST CODE **02138020**

MAY/JUNE 2016

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Boltzmann's constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Density of water	ρ_w	=	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	C_w	=	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of ice	L_f	=	$3.34 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporization of water	L_v	=	$2.26 \times 10^6 \text{ J kg}^{-1}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$
Molar gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Stefan-Boltzmann's constant	σ	=	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Speed of light in free space (vacuum)	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Planck's constant	h	=	$6.626 \times 10^{-34} \text{ J s}$
Triple point temperature	T_{tr}	=	273.16 K
1 tonne	t	=	1000 kg

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided.

1. (a) (i) Define the term 'acceleration' and state its units.

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[1 mark]

- (ii) State Newton's second law of motion in words AND the associated equation.

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[3 marks]

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- (b) A car is travelling along a level road at a speed of 15 m s^{-1} towards a set of traffic lights when the light turns red. In 0.5 seconds after seeing the light, the driver applies the brakes and brings the car to a stop at the traffic lights. Table 1 shows how the speed of the car changes from the time the driver sees that the traffic lights turn red.

TABLE 1

Time/s	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5
Speed/ m s^{-1}	15.0	15.0	12.5	10.0	7.5	5.0	2.5	0.0

- (i) On the grid provided in Figure 1 (page 7), plot a graph of speed against time. [5 marks]

- (ii) Suggest a reason why the velocity remains unchanged for the first 0.5 seconds after the light turns red.

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[1 mark]

- (iii) What feature of the graph shows that the car's deceleration was uniform?

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[1 mark]

- (iv) Use your graph to determine the distance that the car travelled after the lights turned red to when the car stopped.

[4 marks]

Total 15 marks

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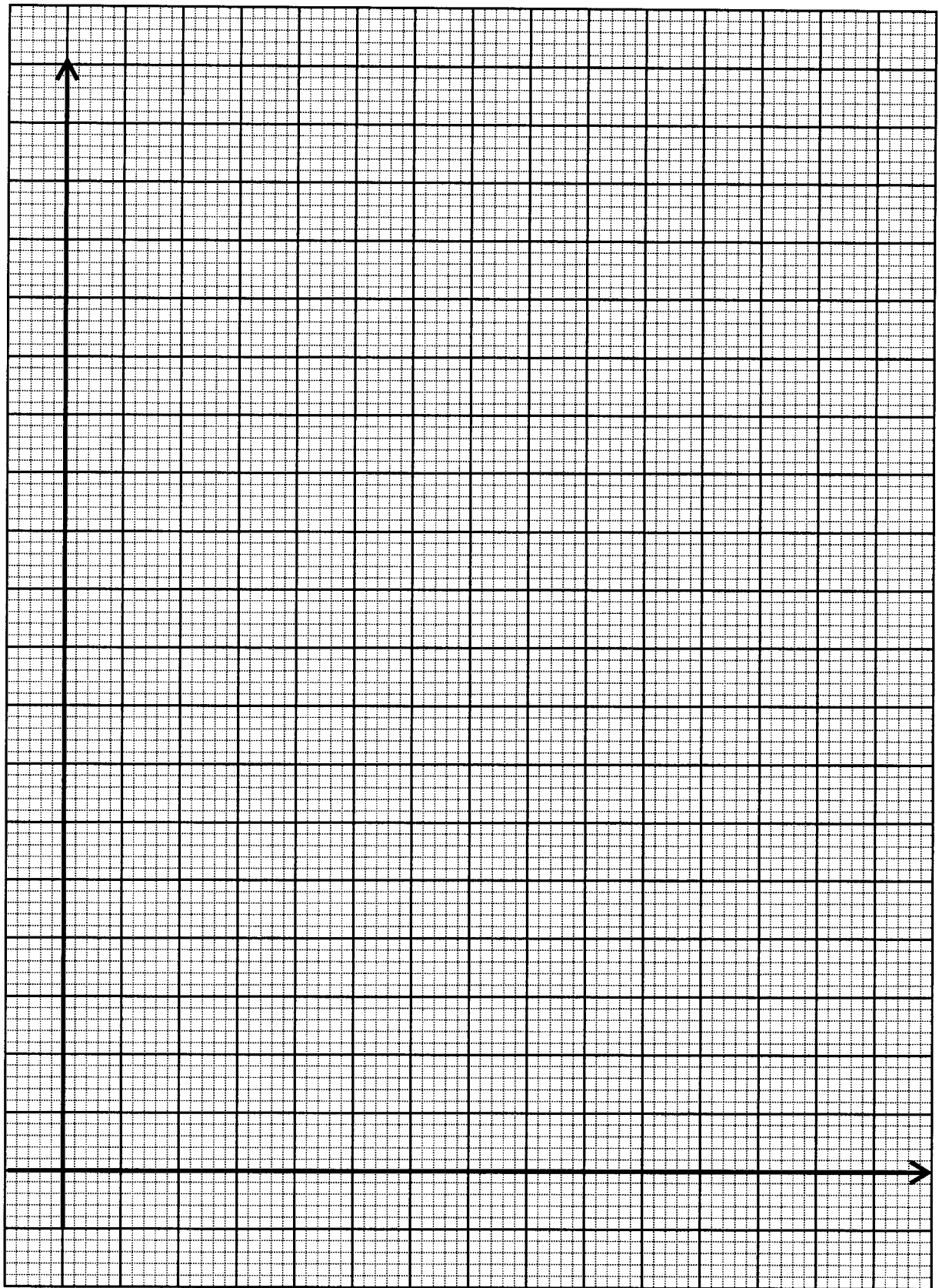


Figure 1. Speed versus time

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2. (a) A transverse progressive wave travels along a stretched string from left to right. The shape of part of the string at a particular instant is shown in Figure 2. The frequency of the wave is 15 Hz.

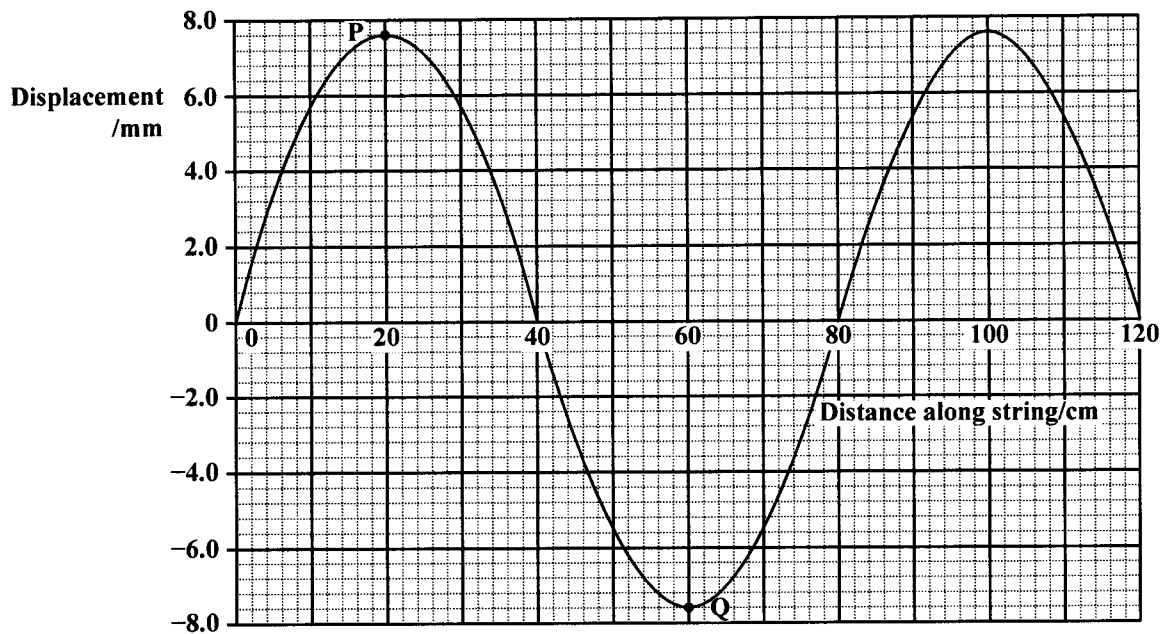


Figure 2. Transverse progressive wave along string

Use Figure 2 to determine, for this wave, its

- (i) amplitude

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 [1 mark]

- (ii) phase difference between Points P and Q on the string

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 [1 mark]

- (iii) speed.

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 [2 marks]

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- (b) Another stretched string is used to form a stationary wave. A part of this wave at a particular instant is shown in Figure 3. The points on the string are at their maximum displacement.

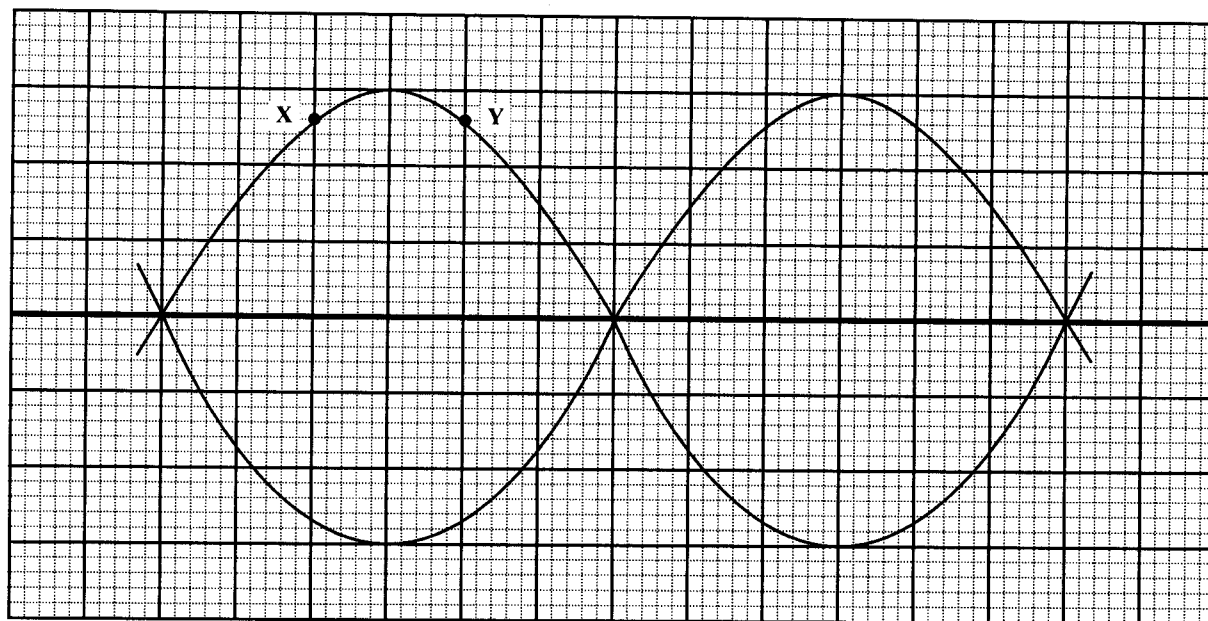


Figure 3. Stationary wave

- (i) State the phase difference in the motion of the points on the string labelled X and Y.

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[1 mark]

- (ii) Distinguish between the terms 'antinode' and 'node' when used to describe stationary waves on a string.

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[2 marks]

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- (iii) State the number of antinodes shown on Figure 3.

[1 mark]

- (c) Table 2 shows how the wavelength produced on a stretched string changes as the wave speed is varied.

TABLE 2

Wave Speed $v/\text{m s}^{-1}$	Wavelength $\lambda \text{ m}$
05	0.06
10	0.12
15	0.18
20	0.24
25	0.30
30	0.36
35	0.42
40	0.48

- (i) On the grid provided in Figure 4 (page 11), plot a suitable graph to represent the data. [5 marks]
- (ii) Hence, determine the frequency of the standing wave.

[2 marks]

Total 15 marks

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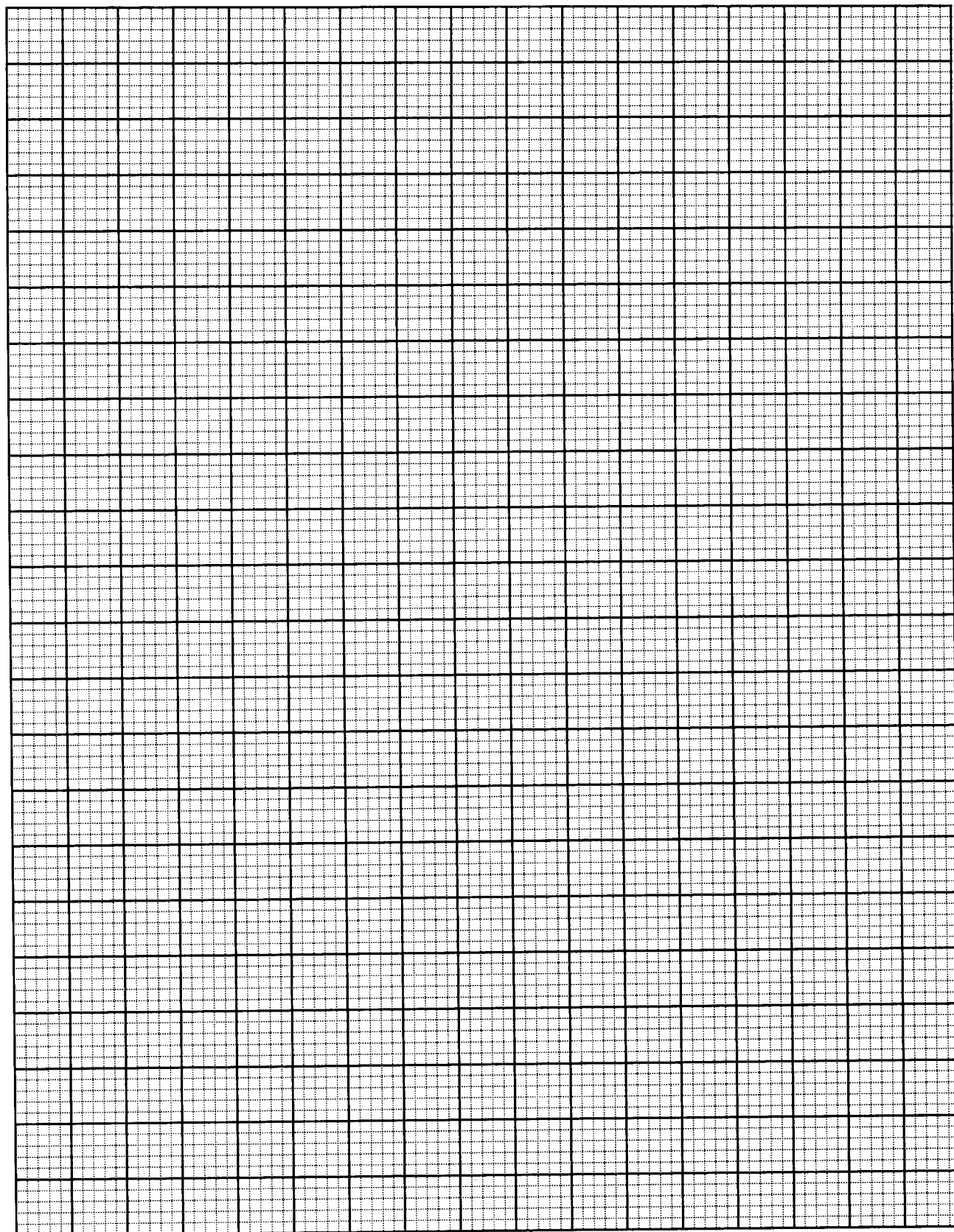


Figure 4. Wavelength versus wave speed

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3. (a) After a lesson on Hooke's law, a student is asked to measure the mass of a rock sample using a steel spring, standard masses and a metre rule. The student measured the unstretched length of the spring and then set up the arrangement shown in Figure 5.

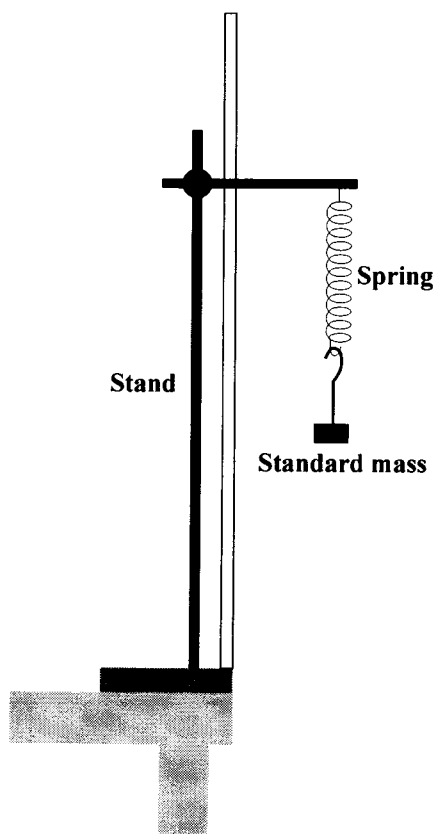


Figure 5. Arrangement for measuring the mass of a rock sample

- (i) State Hooke's law.

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[2 marks]



- (ii) Describe how this arrangement could be used to determine the spring constant and hence, the mass of the rock sample.

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[6 marks]



- (b) A spring manufacturer tests the properties of a spring by measuring the load applied each time the extension is increased. The graph of load versus extension is shown in Figure 6.

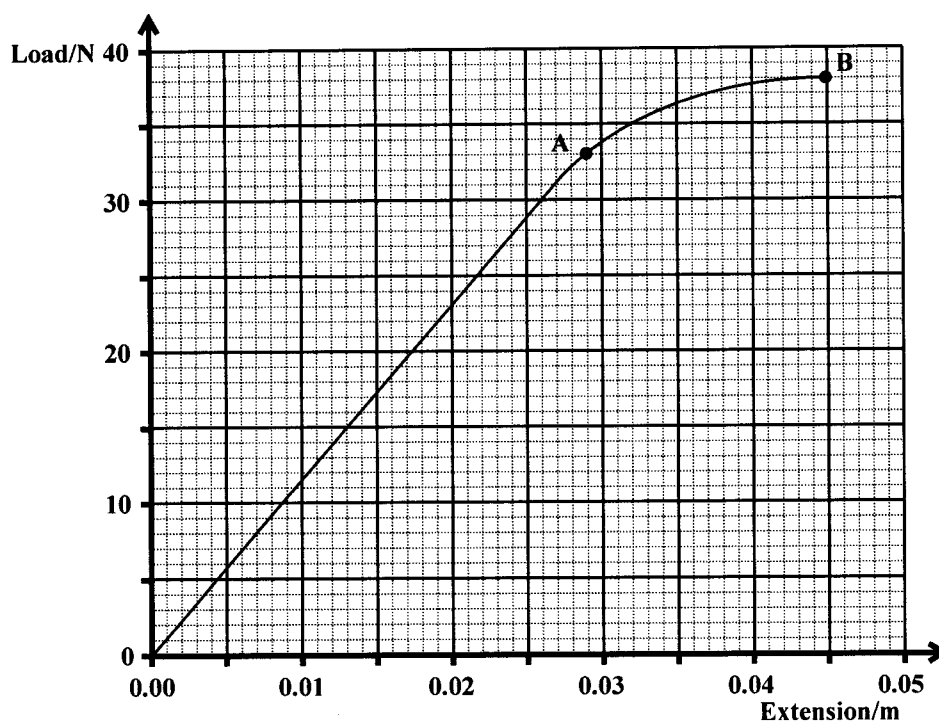


Figure 6. Load versus extension

- (i) Use the graph to find the work done in extending the spring up to Point B.

[5 marks]

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- (ii) Beyond Point A, the spring undergoes inelastic deformation. Explain the term 'inelastic deformation'.

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[2 marks]

Total 15 marks

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided.

4. (a) By referring to Figure 7, explain the origin of upthrust and how upthrust determines whether an object will float or sink.

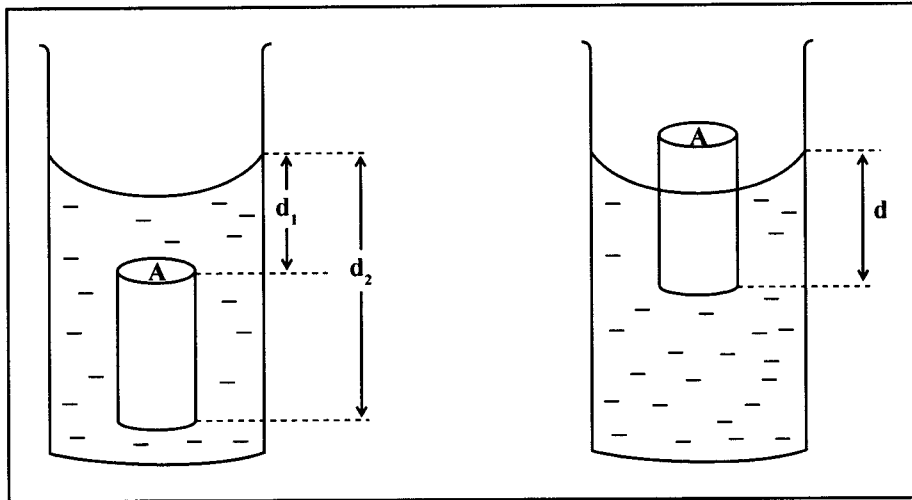


Figure 7. Upthrust

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[5 marks]

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- (b) In the experiment illustrated in Figure 8, it was found that the weight of the stone in air is 1.0 N and the weight when the stone was totally immersed in water was 0.75 N.

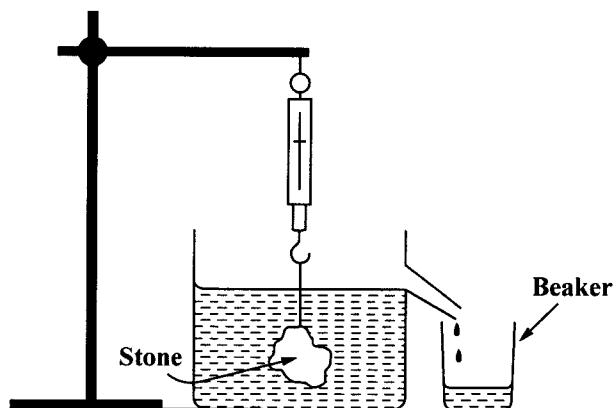


Figure 8. Weight of stone

- (i) Determine the weight of water that would be collected in the beaker.

[1 mark]

- (ii) For (b) (i), calculate the volume of water displaced and hence, determine the density of the stone.

[3 marks]

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- (c) In the experiment shown in Figure 9, it was found that the reading on the spring balance changed from 2.1 N to 1.8 N as the metal ornament was lowered into the measuring cylinder containing water. The height of water in the cylinder rose from the 50 cm³ mark to the 80 cm³ mark. When the cylinder contained corn oil instead of water, it was found that the height of oil rose by the same amount (as when in water), but the reading on the spring balance changed from 2.1 N to 1.87 N.

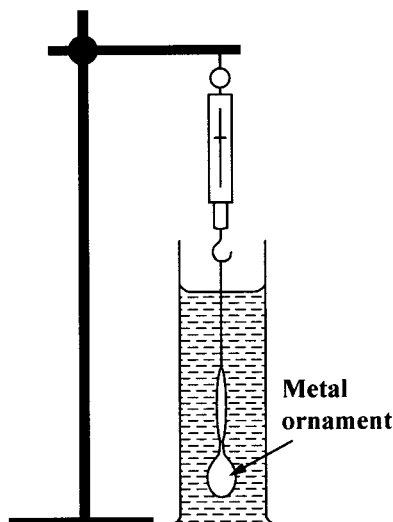


Figure 9. Spring balance

- (i) State the reason why the height of liquid in the measuring cylinder rose by the same amount.

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[1 mark]

- (ii) Outline the reason why the reading on the spring balance was lower when the ornament was immersed in water than when it was immersed in corn oil.

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[1 mark]

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- (iii) Calculate the density of the corn oil.

[4 marks]

Total 15 marks



5. (a) Briefly describe a laboratory experiment to prove Snell's law. Indicate any precautionary measures that must be taken during this experiment.

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[7 marks]

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- (b) Three transparent glass blocks, A, B and C, are arranged (not to scale) as shown in Figure 10. Each glass block has a different refractive index.

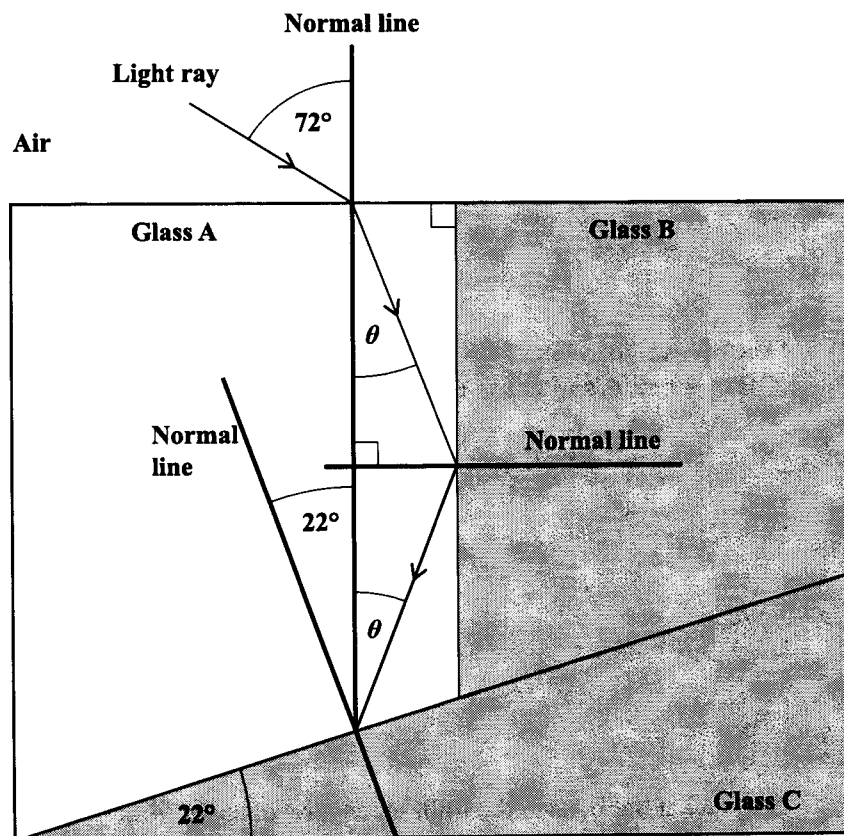


Figure 10. Arrangement of transparent glass blocks

- (i) Calculate the speed of light in Glass Block A.

(Refractive index of Glass Block A = 1.80.)

[2 marks]

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- (ii) Show that the angle θ in the figure is approximately 30 degrees.

[2 marks]

- (iii) The refractive index of Glass Block C is 1.40. Calculate the critical angle between Glass Block A and Glass Block C.

[2 marks]

- (iv) State what happens to the light when it reaches the boundary between Glass Block A and Glass Block C. Justify your answer.

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[2 marks]

Total 15 marks

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- [6 marks]**

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- (b) A rectangular glass window in an office building has dimensions 2.4 metres by 2 metres by 3 centimetres. The temperature of the outer surface of the window is 35 °C and the temperature of the inner surface is 26 °C.

(Assume the thermal conductivity of glass is 0.96 W/m/K.)

Calculate the rate of heat conduction through the glass.

[5 marks]

- (c) The temperature in the room is maintained at 26 °C by an air conditioning unit. If the air conditioning unit fails, determine how long it will take for the temperature of the room and its contents to increase by 3 °C.

(Assume that heat conduction is the only source of heat input to the room; the average heat capacity of the room and its contents is $5.7 \times 10^5 \text{ J/°C}$.)

[4 marks]

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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